



Society of Petroleum Engineers

**SPE-229026-MS**

## **Well Interception to Plug and Abandonment of Inaccessible Wellbore**

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/228859-MS](https://doi.org/10.2118/228859-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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### **Abstract**

An operator in the State of Western Australia (WA) needed to decommission a legacy production well to comply with State regulations and allow for future repurposing of the field for produced water disposal. For one specific well where direct access to the wellbore was unavailable, an intercept well was drilled to carry out the Plug & Abandonment (P&A) operation. Since a precise interception was needed to install well barriers above reservoir depth, a fit-for-purpose survey and ranging program was developed to achieve the interception well objectives. The survey and ranging program were a combination of advanced wellbore survey processing, continuous Passive Magnetic Ranging (PMR) and Active Magnetic Ranging (AMR).

PMR operates by estimating the magnetic field naturally generated by the steel casing within the original wellbore. Unlike AMR, which requires the active generation of a magnetic field, PMR leverages the intrinsic magnetic properties of the well's steel casing. The strength of the magnetic field detected by the magnetometers decreases with distance from the source. By measuring the three-axis magnitudes of the magnetic field at different points as the drill string moves, the system can estimate the distance to the nearby well. The direction to the nearby casing or fish is calculated based on the orientation of the magnetic interference field lines detected by the magnetometers. Multiple measurements are taken from different positions as the tool moves. By combining these measurements, the system can estimate the direction and orientation of the nearby casing or fish relative to the drilling tool.

The combination of PMR and AMR techniques resulted in successful interception on the first attempt and ensured that the original well was securely plugged and abandoned in compliance with the State's regulations. Both PMR and AMR techniques played crucial roles in achieving a precise interception of the original wellbore. The successful application of PMR allowed for the elimination of multiple AMR runs, resulting in significant rig time savings. Traditional AMR available at the time of the project required the Bottom Hole Assembly (BHA) to be pulled out of the hole to be able to run the wireline AMR tool. While the PMR tool is part of the BHA and ranging is done continuously while drilling and rotating the drill-string. The successful correlation of continuous PMR with AMR data reinforced the reliability of PMR, which can be confidently used in other applications such as anti-collision applications, where PMR alone

is typically sufficient, or as an auxiliary method for drilling relief wells, where combination of active and passive ranging techniques is typically advised.

This is the first deployment of next generation measurement-while-drilling technology to perform continuous PMR while drilling and rotating to successfully intercept the target well and complete the P&A operation. The successful correlation of PMR with AMR data reinforced the importance of continuous PMR, which can be applied in other applications such as well collision avoidance applications or relief wells.

## Introduction

An Interception Well (IW) was designed to re-establish access to the Target Well (TW) and allow for installation of reservoir barriers such that the well can be permanently decommissioned. Previous attempts to permanently decommission TW using conventional methods were unsuccessful due to deteriorated well conditions resulting in a loss of access to the deeper section of the well. As a result, TW was suspended with the wellhead in place and shallow (permanent) barriers installed.

The well design for IW included a pre-installed 20" conductor, 13-3/8" surface casing and a 12-1/4" intercept open hole size, milling the window in TW 9-5/8" casing with an 11-7/8" mill. Provision had also been made for contingent 9-5/8" intermediate casing and 8-1/2" intercept open hole size, milling the window in TW 9-5/8" casing with an 8-1/8" mill. The intermediate casing was referred as a Protective Casing string in the event of wellbore instability or other open hole problems that may have required casing off.

The IW trajectory was designed to intercept TW through the 9-5/8" production casing. Primary and secondary intercept points were considered, which were above and below the in-situ TW production packer respectively and designed to allow for intercept with either the 11-7/8" or 8-1/8" mill sizes. Upon completion of operations, IW was to be plugged back and suspended at the surface casing shoe to allow for future repurposing or decommissioning as per Business requirements.

## Methodology

### Directional and Ranging Plan

The intervention well was designed to intersect the target well below 1,970 m TW MD, where the target inclination was  $\sim 60^\circ$ , requiring precise azimuth and inclination control to achieve an optimal low 3D incidence angle. The magnetic ranging strategy combined AMR and Continuous Passive Magnetic Ranging (PMR) and consisted of the following phases:

- Locate Phase – An initial AMR wireline run was planned at  $\sim 1,000$  m IW MD, with an expected  $\sim 10$  m separation and modeled active signal  $> 10$  nT.
- Follow Phase – Continuous PMR was used to monitor the target in real time, reducing AMR runs to critical checkpoints.
- Intercept Phase – Additional AMR runs were performed near the intercept depth to fine-tune alignment prior to "soft touching" the target well casing.

### Active Magnetic Ranging

Active Magnetic Ranging has been used for relief and intercept well drilling for a number of years. Traditional AMR method uses a wireline tool consisting of an electric current injection assembly and an array of magnetic, gyroscopic and acceleration sensors (however, AMR-While-Drilling system is now also available to the industry). Electric current is injected into the formation and some of this current collects on an offset casing string. According to the Ampere's law, a magnetic field is generated by that current around the offset casing string and can be measured and analyzed to determine a distance and direction to that casing.

AMR allows injecting an electric current of known frequency into the formation and, as a result, onto the target casing, which in turn enables superior magnetic signal filtering compared to any other existing ranging method. This enables simple and reliable measurements of the direction to the target casing that can be used for distance-to-target triangulation. In addition, a multi-sensor array inside a typical AMR tool enables active magnetic field gradient measurements when in close proximity to the target well casing, providing an additional way to confirm the distance to the target well.

### Passive Magnetic Ranging

Passive Magnetic Ranging (PMR) is a method which assesses the disturbance in the earth's magnetic field caused by the remanent magnetism of a nearby casing, drillstring or fish in an offset wellbore. These calculations are used to estimate the relative location of the source of that magnetic interference with respect to the PMR sensors by modeling the background magnetic field and subtracting it from the interfered magnetic field measurements, thereby estimating the position of the offset wellbore with respect to the reference wellbore. The sensors required for PMR are a typical MWD or EMS tri-axial arrangements consisting of three magnetometers and three accelerometers, having orthogonal input axes. This method is not new and was initially developed in the 1970s for relief well drilling [1] but has been adapted for a number of different applications where knowledge of a nearby offset wellbore is necessary, such as anticollision monitoring, wellbore twinning, complex plug and abandonments and fish bypass.

A typical casing joint exhibits magnetic dipole properties that are similar to a bar magnet, with positive and negative ends. When these joints are connected together, this produces a measurable magnetic interference along the offset string of casing. The magnetic pattern resulting from the connections at each joint can be compared to a series of magnetic monopoles. The field from each such pole can be estimated, resulting in an estimate of a distance and direction to the source of that magnetic interference. The accuracy of such estimation is dependent on the accuracy of the input data that is used to build a background magnetic field model. Such data includes, for example, an IW survey; therefore, for superior accuracy, a while-drilling gyro survey obtained at or near the magnetic sensor position is required.

PMR analysis typically requires that a series of raw magnetic surveys are taken across individual poles in the offset casing string (often from an MWD tool). To model the interference effectively and obtain the magnetic signature of the casing string, these surveys should be closely spaced. Nominal requirements, which can depend on the specific application, are 10-15 surveys approximately 0.5-1.0m apart, allowing to map a passive magnetic anomaly signature in sufficient detail (Figure 1). The analysis of such PMR data set results in a single distance and direction measurement to the source of interference.

In a standard MWD tool, these surveys require to be MWD to be stationery and therefore when trying to range to another wellbore, the process can be extremely time consuming, and the constant recycling of the mud pumps can have a detrimental effect on wellbore condition. It has been demonstrated that the use of Dynamic Definitive Surveys DDS can be very effective in producing Continuous PMR results with minimal impact on rig time [2]. The next generation MWD technology featuring DDS is utilized in this case study. The ability of the next generation MWD tool to take continuous 6-axis dynamic surveys is not the only requirement for a successful real-time PMR job. Additionally, the system should be able to provide sufficient surveying data density and accuracy, requiring while-drilling gyroscopic survey measurements near the magnetic sensors for effective background field modeling and cancelation, an acceptable signal-to-noise ratio, and advanced ranging processing to interpret the data in real-time.

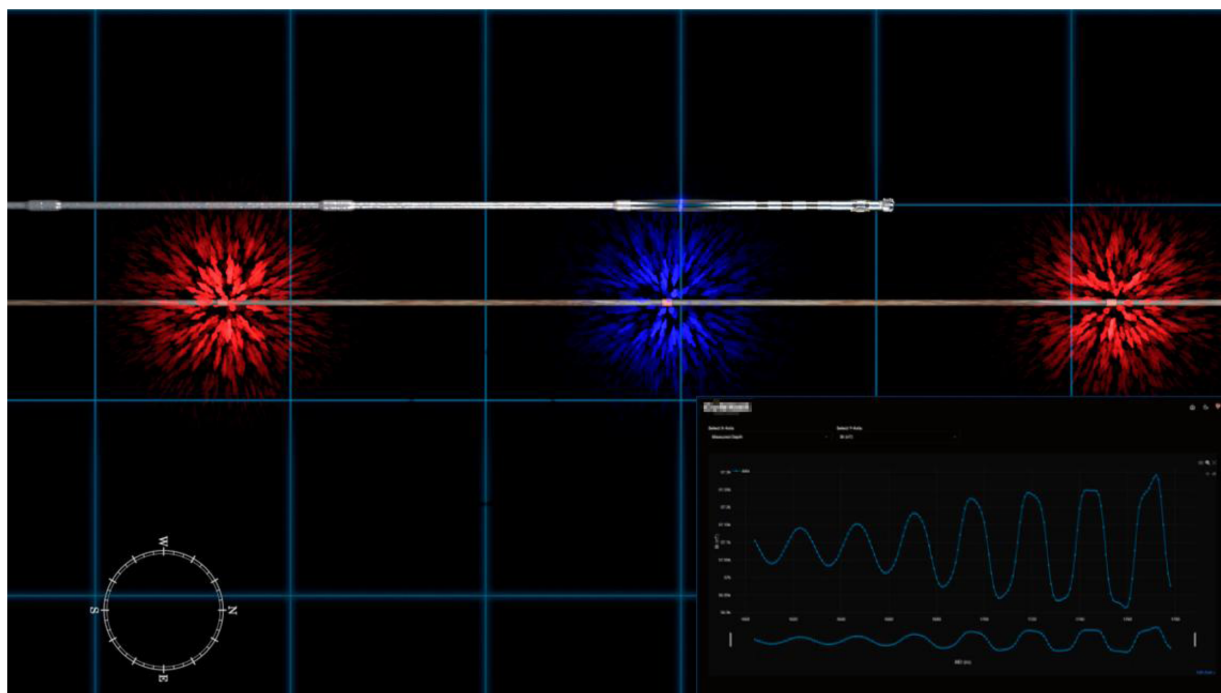


Figure 1—Illustration of Passive Magnetic Ranging data collection.

Following is a list of critical elements to enable successful Continuous PMR operation:

**Definitive Dynamic Surveys DDS.** Details of the DDS design and error model description have been discussed in [3] and [4]. The DDS allows the same or better levels of accuracy to be achieved when drilling, eliminating the need for stationary surveying time. Consequently, a substantial rig time can be saved especially when Passive Magnetic Ranging (PMR) is required to range to the offset well.

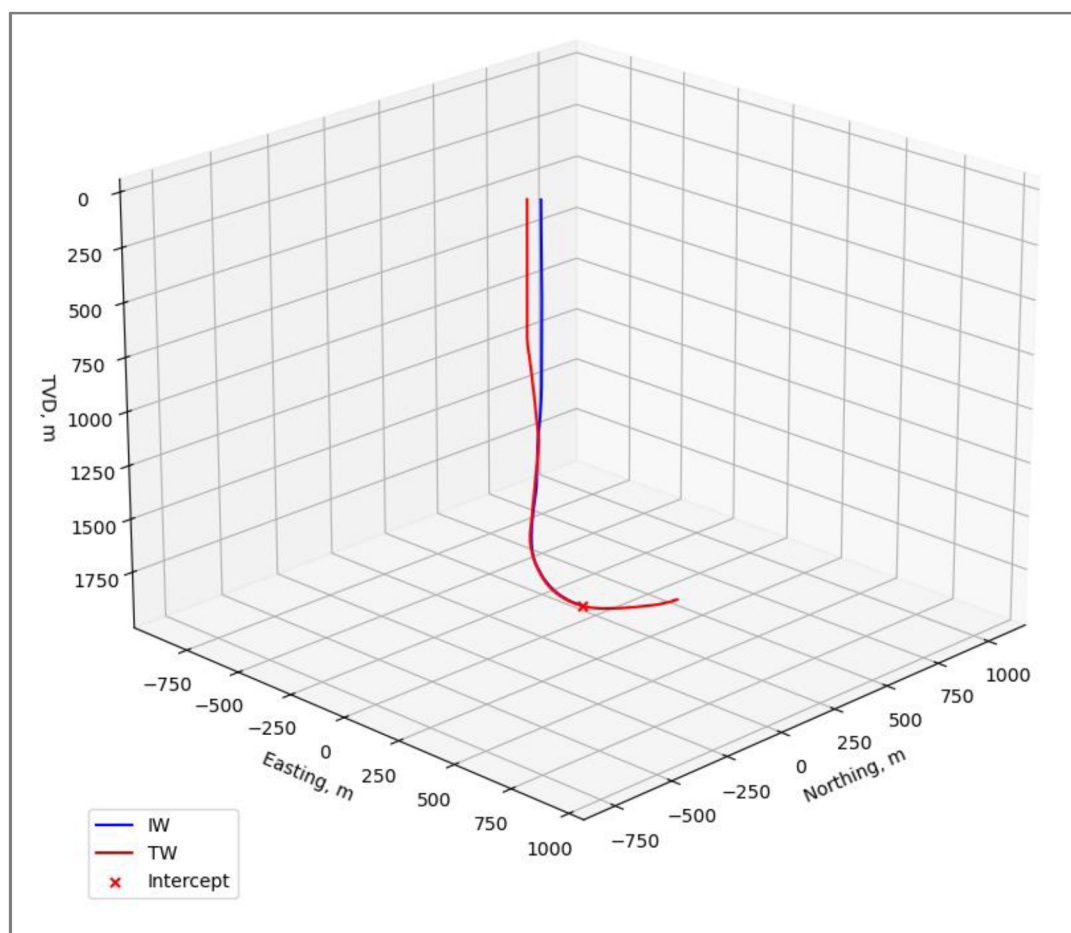
**PMR cloud-based platform.** The processing platform allows the use of several ranging models simultaneously to have more control over the confidence of the ranging results. Priority was given to simplicity to enhance real-time interpretation speed; therefore, "geometrical" methods were used. The new models have evolved from the techniques described in [5] and enable users to perform ranging from a greater distance to the offset well. The models consider expected casing charge, achieving a better fit of radial and axial signals.

**DDS frequency.** The distance between two consecutive surveys to accurately measure the magnetic pole should be within 1 m. The required refresh rate depends on the expected ROP and the requirements of any other downhole measurements. When operating in the special ranging mode, the tool can take a dynamic survey every 30 seconds, providing sufficient ranging data density for ROP's up to 120 meters per hour.

## Execution

Prior to operations, the target well was extensively studied with ranging and intercept execution plan developed well in advance working in close collaboration with the customer. The intercept well was planned to intersect the target well below 1,970m TW MD (measured depth along the target well), where the target well is continuously building angle (Figure 2). The target well inclination at the planned intercept depth was approximately 60°, requiring tight control of the intercept well azimuth and inclination to precisely align with the target from above at the optimal 3D incidence angle.





**Figure 2—Trajectories of the target well (TW) and planned intercept well (IW). The intercept well approaches and follows the target well before the intercept occurs at approximately 1,970m IW MD, at an inclination of approximately 60°.**

Operation began with an extensive site visit involving all stakeholders. This included inspecting the surface location, the offset well, and the rig facilities to assess the placement of the MWD unit and the running of the Wireline (WL). A visit to the SLB base followed, during which active ranging and gyro tools were tested for compatibility with the WL rope socket. Additionally, an interchangeable WL head was developed to work with different tools, minimizing time lost between runs. A dedicated generator was also prepared and tested to inject the high current required for active ranging operations.

Drilling operations commenced in Q2 2024 with the 17 ½" section. The first run was drilled to a depth of 1013 meters, where the initial Active Magnetic Ranging (AMR) run was planned. Following the first active ranging process, the offset well was located at the expected distance and direction. The ranging results were used to adjust the subject well's surveys to align the center-to-center (ct-ct) distance and relative direction with the offset well and update the intercept well plan. The 17 ½" section was completed in two runs, and the 13 3/8" casing was successfully installed.

The 12 ¼" section was initiated using a Rotary Steerable System (RSS) Bottom Hole Assembly (BHA) specially designed with the RSS, MWD and GWD tools as illustrated in Figure 3. The point-the-bit RSS provided on-demand dogleg severity (DLS) capabilities. This was complemented by the latest MWD technology, and high accuracy GWD, which uses solid-state gyro sensors for gyro measurements. This advanced BHA configuration enabled precise adherence to the directional plan, accurate surveys, and Passive Magnetic Ranging (PMR) while drilling. The first PMR processing at 1334 meters showed excellent correlation with the expected proximity and AMR results as shown in Figure 4.

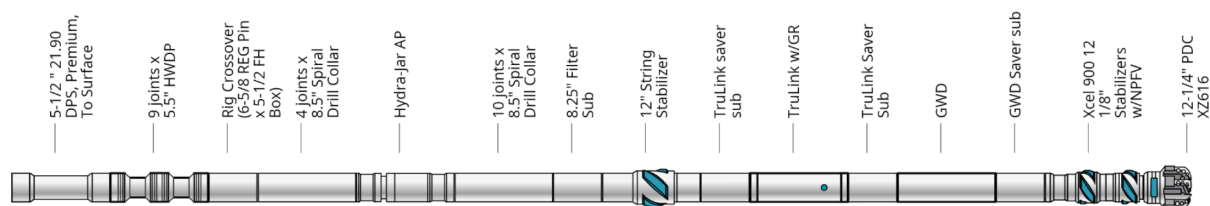


Figure 3—Ranging while drilling Bottom Hole Assembly

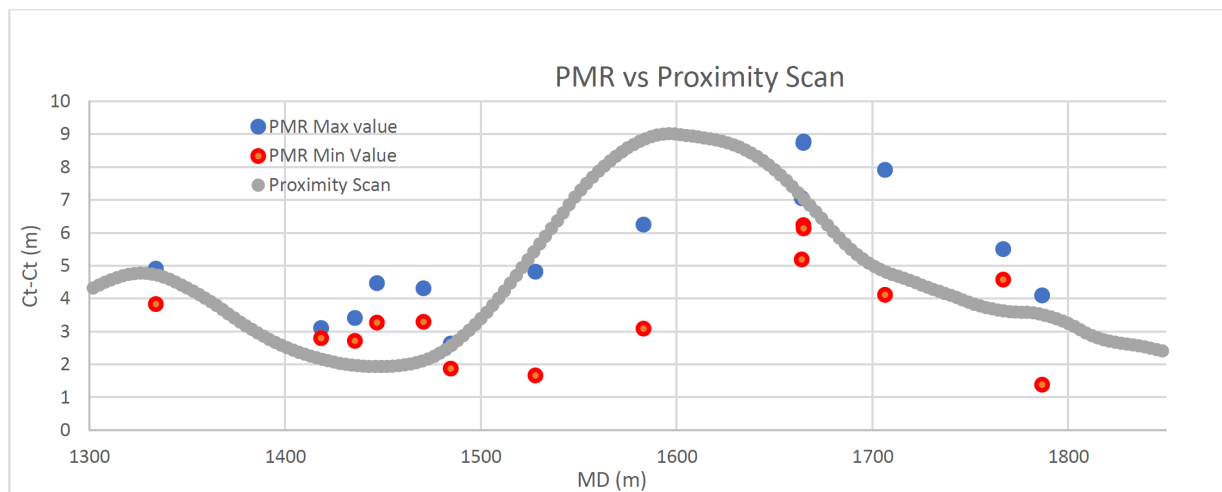


Figure 4—Passive Magnetic Ranging Results

Drilling continued with the RSS BHA and periodic PMR processing, progressively bringing the subject well closer to the offset well. At this stage, the BHA was switched to a motor assembly shown in Figure 5 to enable shorter runs and higher on-demand DLS. Additionally, a dedicated survey BHA was assembled, incorporating a gyro sensor positioned near the bottom to capture highly accurate surveys as close as possible to the offset well as illustrated in Figure 6.

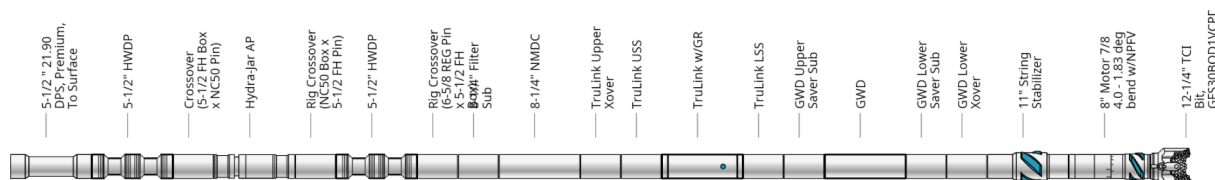


Figure 5—Mud Motor Bottom Hole Assembly

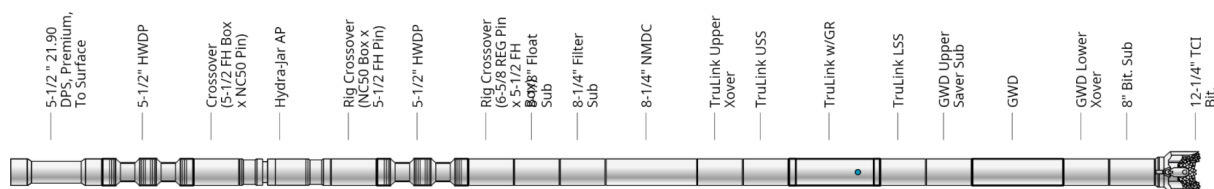


Figure 6—Bottom Hole Assembly with gyro close to bit

This process—combining AMR, PMR, and gyro measurements—was repeated until successful interception of the target well as per the operational plan. The final trajectory was aligned to provide the required toolface-to-target orientation for casing milling operations.

To ensure a high-quality milled window for target re-entry, a specially designed mill was developed to penetrate the 9 5/8" casing. This mill was deployed in a Motor BHA with a 1.83° bend, along with an MWD

tool to orient it to the low-side toolface (TF). This precise orientation enabled the creation of a window with the dimensions and quality required to proceed with the subsequent Plug and Abandonment (P&A) operations.

The milling procedure and window dimensions were carefully planned well in advance, based on the specific P&A requirements for this well. The casing milling was successfully completed in a single run. To ensure the window was fully prepared for the final P&A stage, the assembly was reciprocated through the window to clear all edges and achieve the desired finish.

## Results

Upon contact with the target well, a dedicated milling assembly was run into the intervention well. After confirming contact via noticeable torque, window milling commenced into the target well's 9 $\frac{5}{8}$ " casing in accordance with a specifically designed milling model as illustrated in Figure 7. The successful milling operation was completed as shown in Figure 8. Upon completion of the window milling, the abandonment assembly was run through the window into the target well, and the required cement barrier was set in conformance with local regulations following regulatory approval for the target well's abandonment.

Continuous Passive Magnetic Ranging (PMR) played an important role in the ranging operation, significantly optimizing the process by eliminating the need for additional wireline-conveyed Active Magnetic Ranging (AMR) runs. This efficiency saved approximately 10 days of rig time. Ultimately, the operator successfully decommissioned the well, paving the way for the facility to be prepared for the next project.

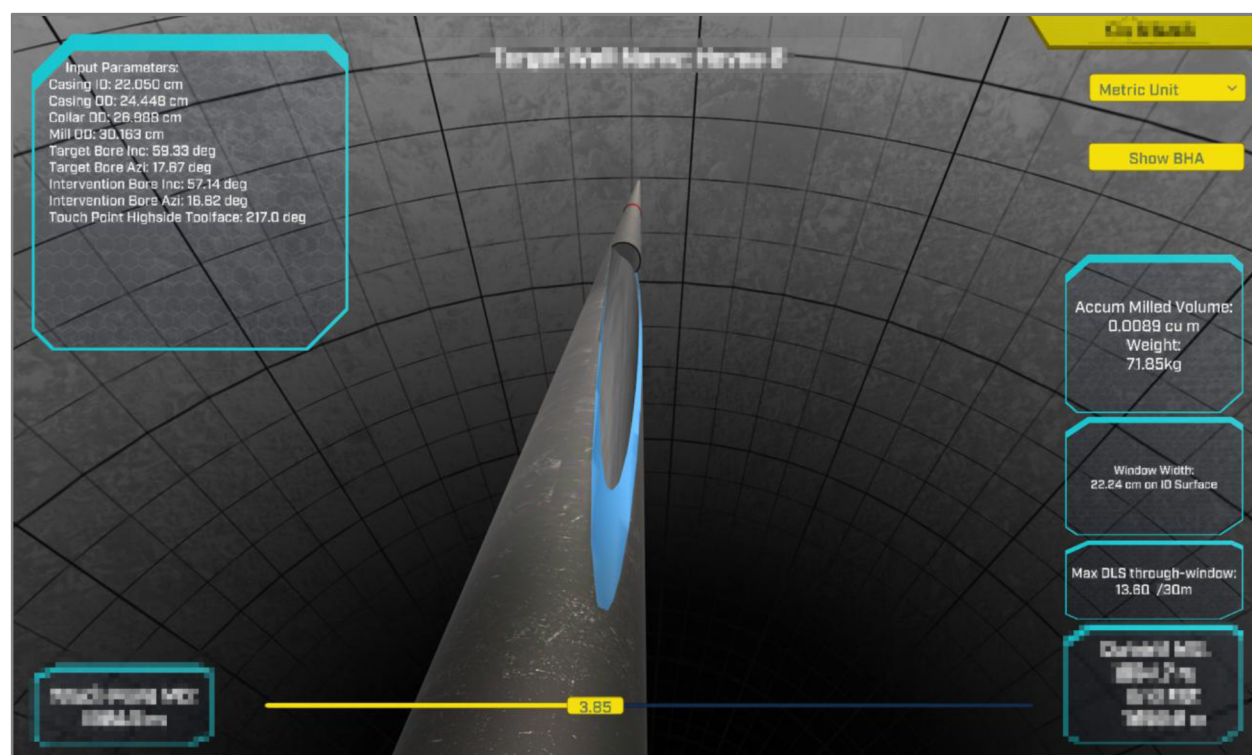


Figure 7—Model used for milling a window into the target well's 9 $\frac{5}{8}$ " casing. The model served as a guide to ensure optimal window dimensions, enabling the successful re-entry of the abandonment assembly into the target well casing





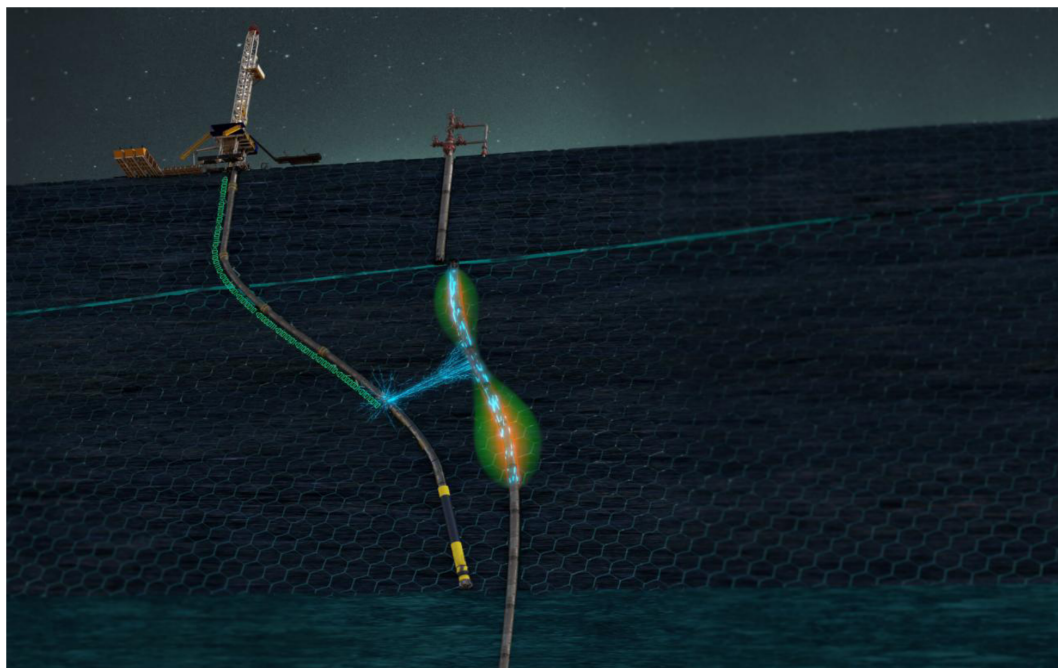
**Figure 8—(a) The brand-new 11 7/8" mill being picked up prior to deployment into the intervention well for milling a window in the target well casing. (b) After completing the window milling, the same mill is shown alongside samples of metal swarf collected from the ditch magnets.**

## Conclusions

The interception and re-entering the TW for P&A operation was completed successfully and per the plan. The integration of Continuous PMR into the AMR ranging schedule has proven to be beneficial. The correlation of PMR with AMR data reinforced the efficiency of PMR, which proved that it can be effectively used in other applications such as well collision avoidance or to significantly augment and optimize ranging strategy in relief well and complex abandonment applications.

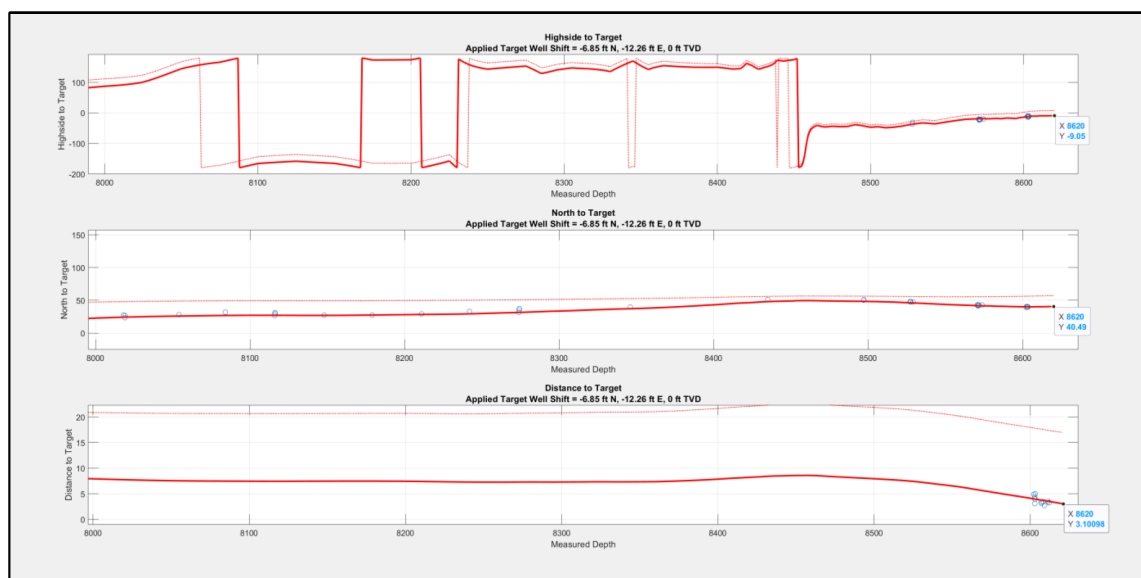
## New Tech Update

Since the time of the operations on this project, a new AMR-While-Drilling system has been introduced to the industry [6]. The system is based on the use of powered drill pipe, enabling multiple-times-enhanced formation current injection and autonomous AMR measurement and processing downhole, with wireless mud-pulse transmission of results to surface (a 3D illustration of the new system in action is shown in Figure 9).



**Figure 9—Illustration of the new AMR-While-Drilling system.**

The most recent deployment of the new system in North America in August 2025, using powered pipe, proved its efficiency in oil-based mud and salt formations. It successfully intercepted a very narrow intercept window of a target well (AMR-While-Drilling ranging data using powered-pipe is shown in Figure 10), facilitated immediate kill operations, which were followed by milling a window in a target casing allowing re-entry into the target well with a tubing stinger for permanent abandonment.



**Figure 10—AMR-While-Drilling ranging data from the August 2025 North America deployment, obtained with powered drill pipe current injection in oil-based mud prior to the intercept of the target well.**

The new AMR-While-Drilling system enables drilling an intercept well without BHA trips, increases range through superior current injection capability and improves ranging accuracy by being perfectly aligned with the intercept wellbore axis. In addition, it allows both gyro surveying and active ranging from the same BHA while drilling. These capabilities enable faster blowout response and safer relief well drilling operations by keeping the BHA and drill string in the hole at all times, thereby allowing immediate well-control actions in unstable downhole conditions.

## Acknowledgement

The authors thank SLB, and Gunnar Energy Services for permission to publish this case study and the SLB WCM, SLB WEC Ranging & Interception team and Gunnar Energy Services team who ensured the success of this operation.

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